FORENSICS IN DIGITAL SECURITY (FDS)

Digital Forensic Tools II

OBJECTIVES

- Describe types of different graphics formats
- Explain types of data compression
- Explain how to locate and recover graphics files
- Describe identifying and reconstruction of graphic images
- Describe how to identify unknown file formats
- Explain copyright issues with graphics
- Describe the use of steganography

RECOGNIZING A GRAPHICS FILE

What is a graphics file?

Any File

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Contains

Bitmap (or Raster) images
- Collection of dots

Vector graphics
- Based on mathematical instructions

Metafile graphics
- Combination of bitmap and vector

Examples

- Digital photographs
- Line art
- Three-dimensional images
- Scanned replicas of printed pictures, etc.

•Types of programs

- Graphics editors
- Image viewers

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

BITMAP AND RASTER IMAGES

Bitmap and raster images are stored as series of tiny dots called pixels.

Difference between bitmap and raster?

Bitmap images

- Grids of individual pixels

Raster images

- Pixels are stored in rows
- Better for printing

Factors affecting Image quality

- Screen resolution
- Software
- Number of color bits used per pixel

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

VECTOR IMAGES

Characteristics

- Lines instead of dots
- Store only calculations for drawing lines and shapes
- Smaller size
- Preserve quality when image is enlarged

•Example of vector illustration software

- CorelDraw
- Adobe Illustrator

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

METAFILE IMAGES

Combination of bitmap and vector images

Share advantages and disadvantages of both bitmap and vector types

When enlarged, bitmap part loses quality

Examples

Scanned photo (bitmap) with text (vector)

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

SOME GRAPHICS FILE FORMATS

Graphics Formats

Non- standard

- Targa (.tga)
- Raster Transfer Language (.rtl)
- Adobe Photoshop (.psd) and Illustrator (.ai)
- Freehand (.fh9)
- Scalable Vector Graphics (.svg)
- Paintbrush (.pcx)
- Etc.

Standard

Bitmap

- Graphic Interchange Format (.gif)
- Joint Photographic Experts Group (.jpeg, .jpg)
- Tagged Image File Format (.tiff, .tif)
- Window Bitmap (.bmp),
- Etc.

Vector

- Hewlett Packard Graphics Language (.hpgl)
- Autocad (.dxf),
- Etc.

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

DIGITAL CAMERA FILE FORMATS

Anyone can create digital images

Witnesses or suspects can create their own digital photos

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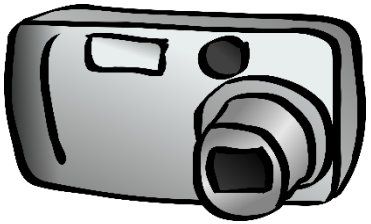
Can the images be trusted?



Digital Camera

Camera sensors record pixels

Stores pixels in camera's memory card

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Recorded as

Raw file format

- Minimally processed data from sensors
- Typically found on many higher-end digital cameras
- Maintains best picture quality

Raw file format

Examine?

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

DIGITAL CAMERA FILE FORMATS

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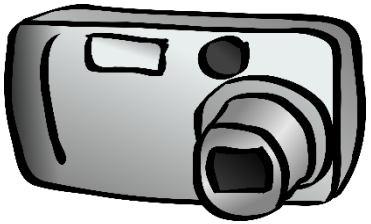

Different Digital Cameras

Stores

Different Proprietary Raw File Formats

Biggest Disadvantage

Need to convert raw image data to another format?

Process of converting raw to another format

Demosaicing

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Raw file format

to

Standard format e.g. JPEG, TIF

Examine?

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

DIGITAL CAMERA FILE FORMATS

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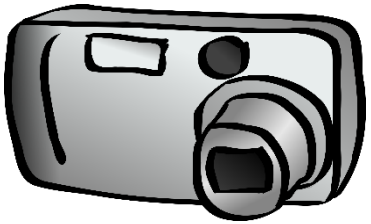

Different Digital Cameras

Stores

Exchangeable Image File (EXIF)

Developed by JEIDA as a standard for storing metadata in JPEG and TIFF files

EXIF format collects metadata and stores metadata at the beginning of the file

Extract metadata as evidence with tools such as ProDiscover and Exif Reader

Investigators can learn more about the digital camera and the environment in which pictures were taken

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

COMPARING DIGITAL CAMERA FILE FORMATS

EXIF JPEG **Standard JPEG**

Sawtoothmt.jpg

Sawtoothmtn.jpg

Figure 10-1 Identical EXIF and JPEG pictures

Figure 10-1 Identitcal EXIF and JPEG pictures. Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

DIFFERENCE DIGITAL CAMERA FILE FORMATS (EXIF JPEG VS JPEG)

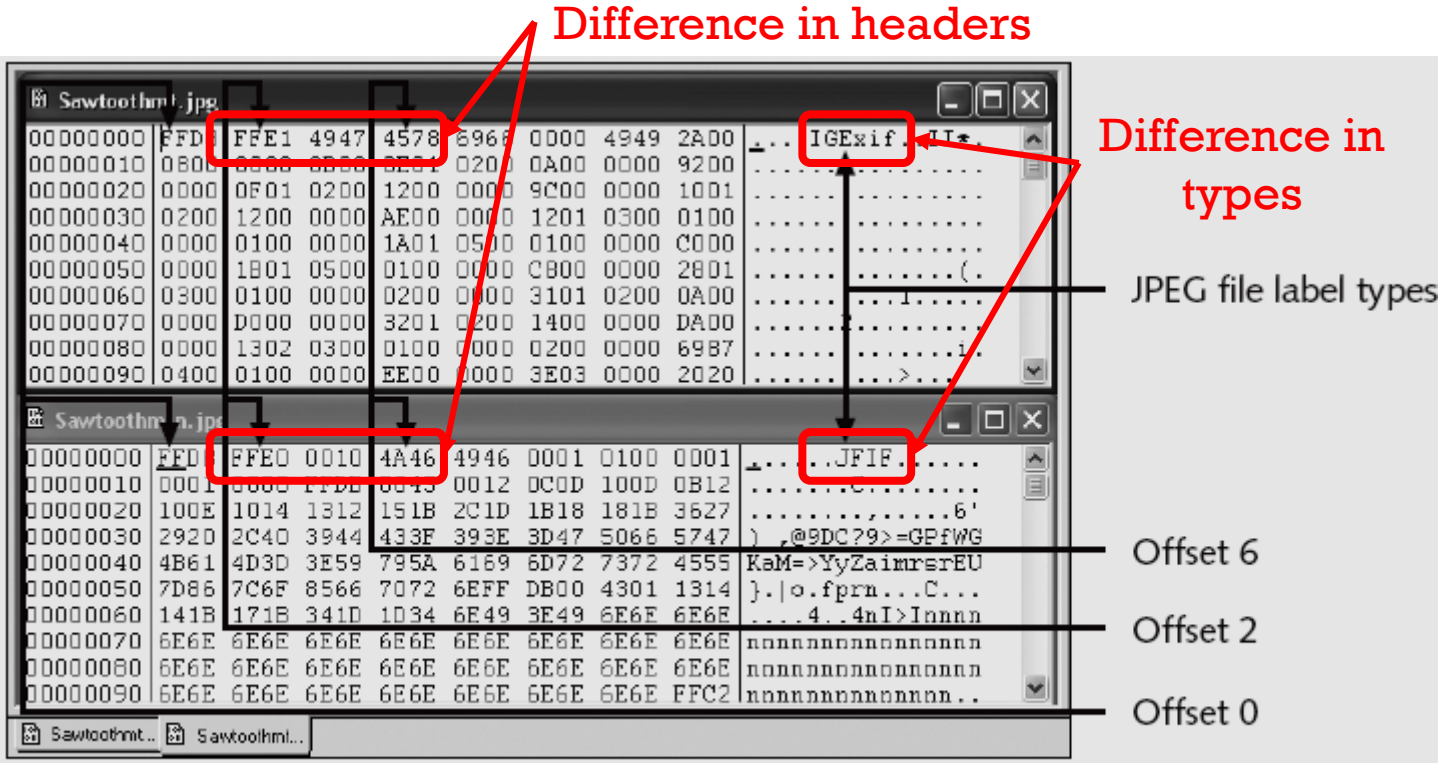

Figure 10-2 Differences in EXIF and JPEG file header information

Figure 10-2 Difference in EXIF and JPEG File Header Information. Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

DIGITAL CAMERA FILE FORMATS (JPEG EOI MARKER)

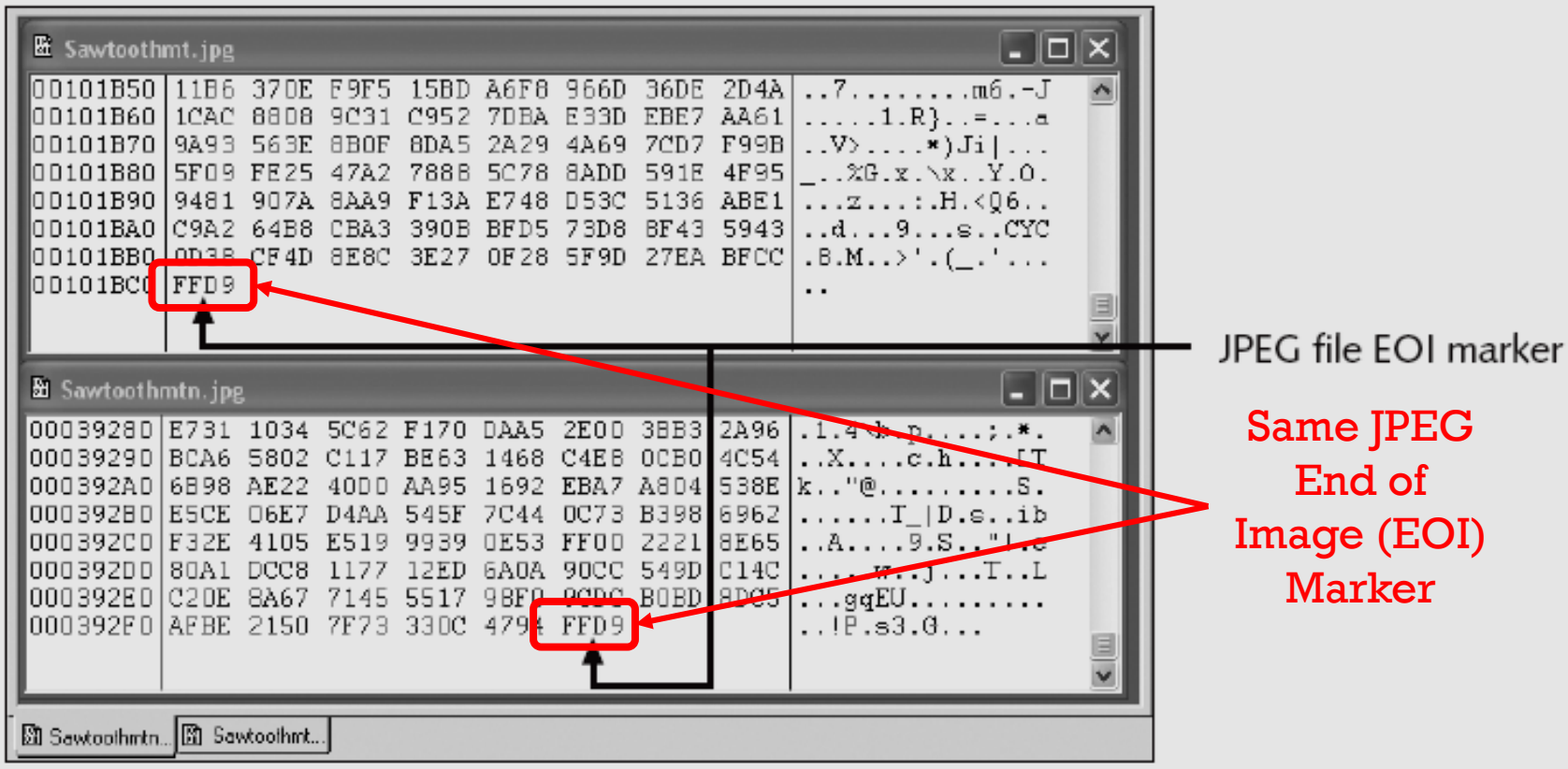

Figure 10-3 EOI marker FFD9 for all JPEG files

Figure 10-3 EOI Marker FFD9 for all JPEG files. Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

DIGITAL CAMERA FILE FORMATS (EXIF METADATA)

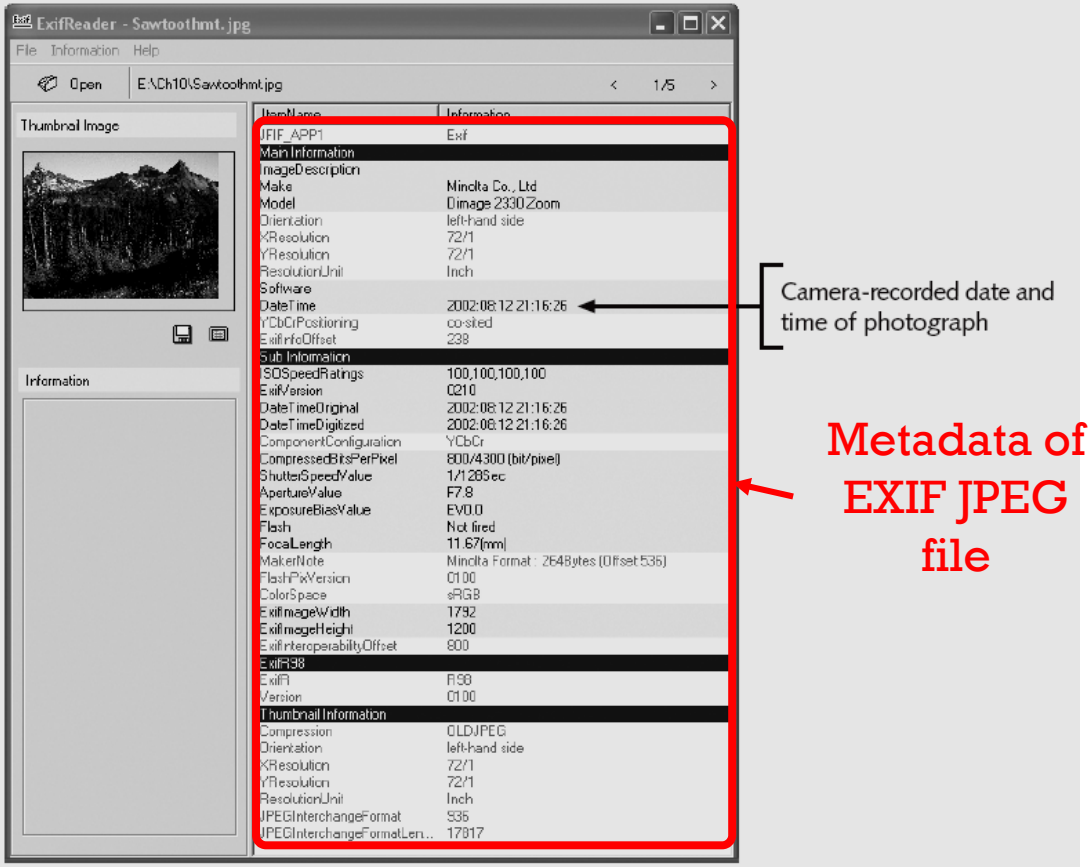

Figure 10-4 Exif Reader displaying metadata from an EXIF JPEG file

Figure 10-4 EXIF Reader displaying metadata from an EXIF JPEG file. Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

DATA COMPRESSION

Some image formats compress their data . e.g . GIF, JPEG, PNG
Others, like BMP, do not compress their data.

Data Compression → Coding of data from a larger to a smaller form

Lossless Compression

- Reduces file size without removing data
- Based on Huffman or Lempel-Ziv-Welch coding
  - For redundant bits of data
- Utilities: WinZip, PKZip, StuffIt, and FreeZip

Lossy Compression

- Permanently discards bits of information
- **Vector quantization (VQ)**
  - Determines what data to discard based on vectors in the graphics file
- Utility: Lzip

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

LOCATE AND RECOVER GRAPHICS FILES

Tools to locate and recover graphics

Operating system tools	Computer forensics tools

• Time consuming
• Results are difficult to verify
	• Image headers
	  • Compare them with good header samples
	  • Use header information to create a baseline analysis
	• Reconstruct fragmented image files
	  • Identify data patterns and modified headers

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

IDENTIFY AND RECOVER GRAPHICS FILE FRAGMENTS

Use of computer forensics tools

Help identify image files fragments and put them together

Data Carving

- Salvaging and Recovering all File Fragments
- Carve from Slack Space and Free Space

Repairing Damage Headers

Use good header samples

- Each image file has a unique file header
- e.g. JPEG: FF D8 FF E0 00 10
  - Most JPEG files also include JFIF string

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

EXAMPLE: EMTS IP THEFT

Investigate a possible intellectual property theft by a contract employee of Exotic Mountain Tour Service (EMTS)

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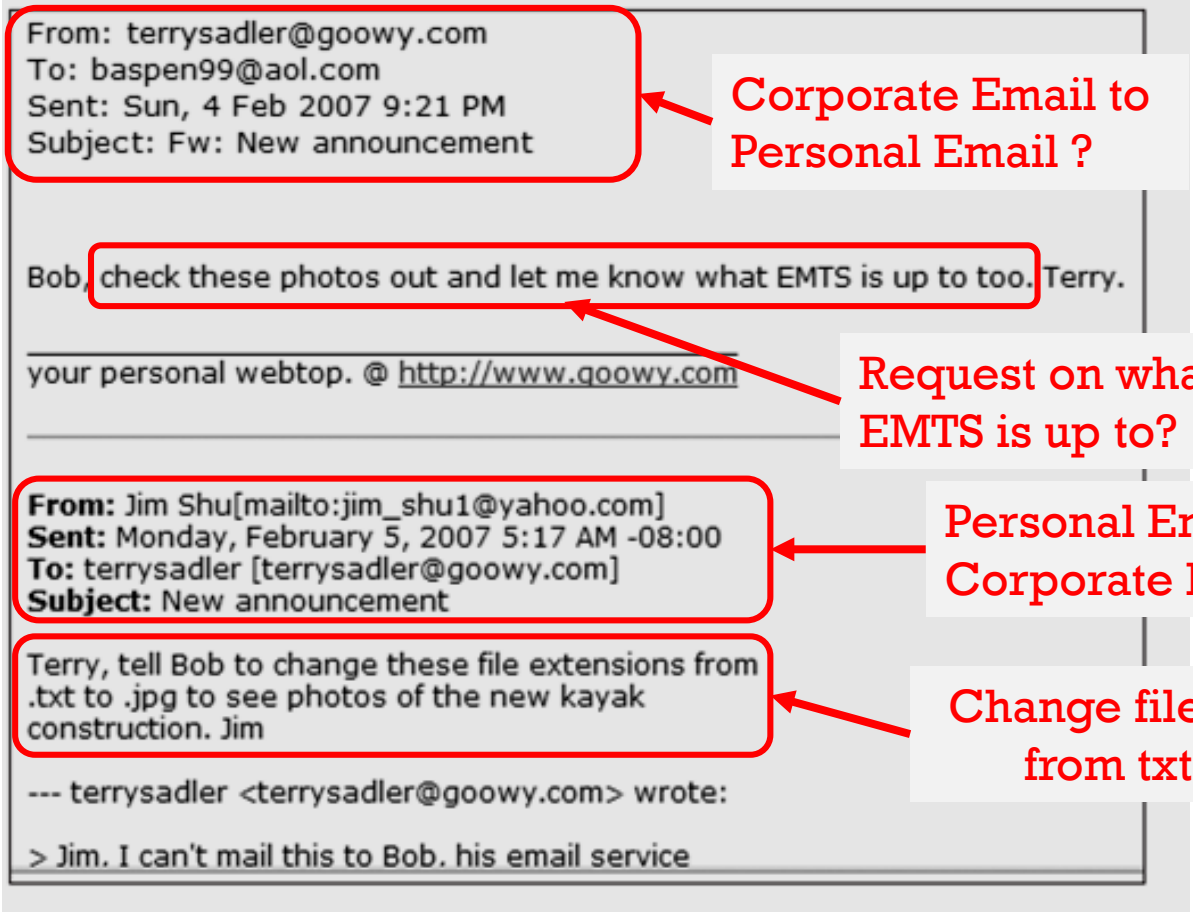
How should the investigation be carried out?

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

EXAMPLE: EMTS IP THEFT (POSSIBLE INDICATORS)



From: terrysadler@goowy.com
To: baspen99@aol.com
Sent: Sun, 4 Feb 2007 9:21 PM
Subject: Fw: New announcement

Bob, check these photos out and let me know what EMTS is up to too. Terry.

your personal webtop. @ http://www.goowy.com

From: Jim Shu[mailto:jim_shu1@yahoo.com]
Sent: Monday, February 5, 2007 5:17 AM -08:00
To: terrysadler [terrysadler@goowy.com]
Subject: New announcement

Terry, tell Bob to change these file extensions from .txt to .jpg to see photos of the new kayak construction. Jim

--- terrysadler <terrysadler@goowy.com> wrote:

> Jim. I can't mail this to Bob. his email service

Corporate Email to Personal Email ?

Request on what EMTS is up to?

Personal Email to Corporate Email?

Change file extension from txt to jpg?

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What are the possible indicators?

Figure 10-5 First intercepted capture of an Email Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

EXAMPLE: EMTS IP THEFT (POSSIBLE INDICATORS)

From: denisesuperbic@hotmail.com
To: baspen99@aol.com
Sent: Sun, 4 Feb 2007 9:29 PM
Subject: RE: New announcement

Can you read the attachments yet? Denise

>From: Jim Shu <jim_shu1@yahoo.com>
>To: terrysadler <terrysadler@goowy.com>
>CC: nautjeriko@lycos.com
>Subject: New announcement
>Date: Sun, 4 Feb 2007 20:57:37 -0800 (PST)
>
>Terry,
>
>I had a tour of the new kayak factory. I think we can
>run with this to the other party interested in
>competing. I smuggled these files out, they are JPEG
>files I edited with my hex editor so that the email
>monitor won't pick up on them. So to view them you
>have to re-edit each file to the proper JPEG header of
>offset 0x FF D8 FF E0 and offset 6 of 4A. Then you
>have to rename them with a .jpg extention to view
>them.
>
>See attached, Bob Aspen I think is working at EMTS he

Email to the same recipient?

Changing of JPEG header and file extension?

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What are the possible indicators?

Figure 10-6 Second intercepted capture of an email Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

EXAMPLE: EMTS IP THEFT (POSSIBLE INDICATORS)

Planning your examination

Searching for and recovering digital photograph evidence

Use ProDiscover to search for and extract (recover) possible evidence of JPEG files	False hits are referred to as **false positives**

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

EXAMPLE: EMTS IP THEFT (SEARCH FOR POTENTIAL EVIDENCE)

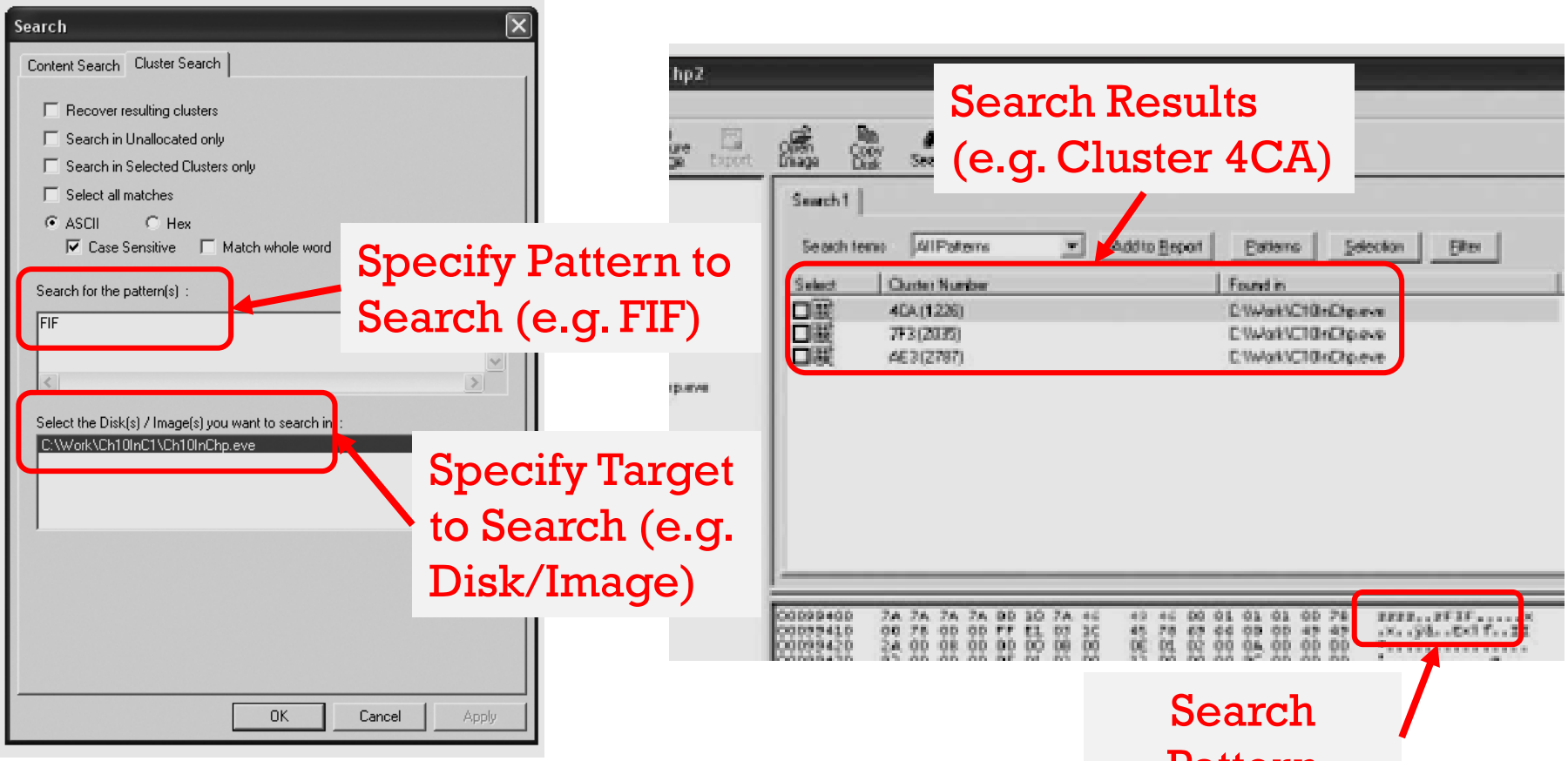
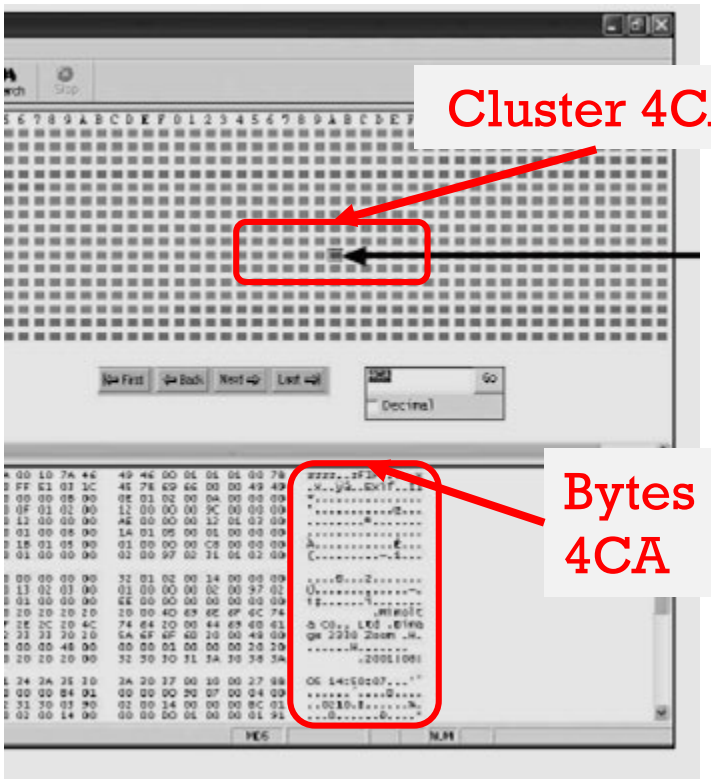


Figure 10-7 Searching clusters in ProDiscover
Adapted from Figure 10-8 Completed search in ProDiscover
Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

EXAMPLE: EMTS IP THEFT (SEARCH FOR POTENTIAL EVIDENCE)

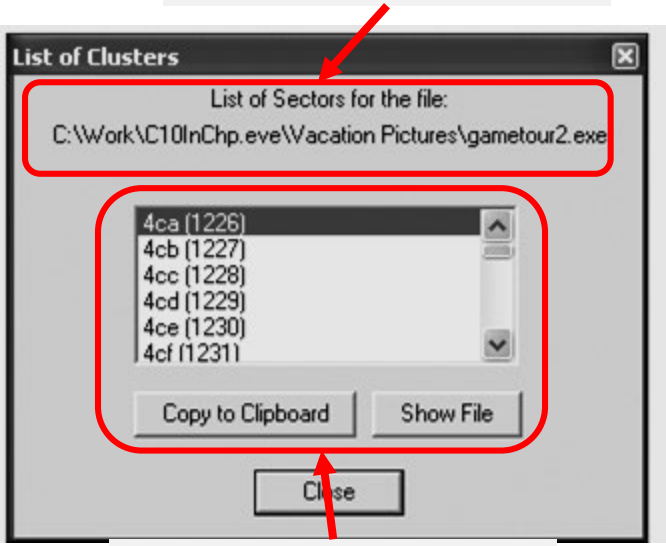


File header was changed to zzzz

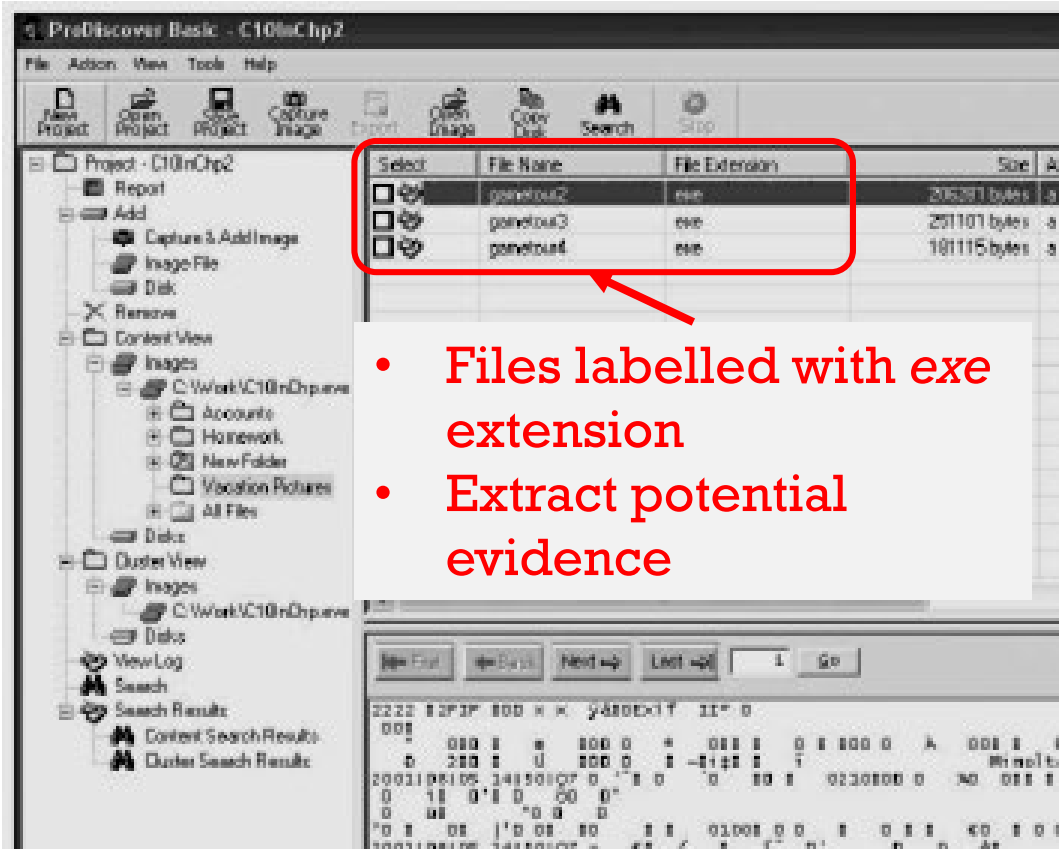
Adapted from Figure 10-9 View cluster use and location at 4CA
Adapted from Figure 10-10 Content of cluster 4CA
Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

EXAMPLE: EMTS IP THEFT (IDENTIFY FOR POTENTIAL EVIDENCE)

Identified file based on cluster 4CA



Sectors of identified file



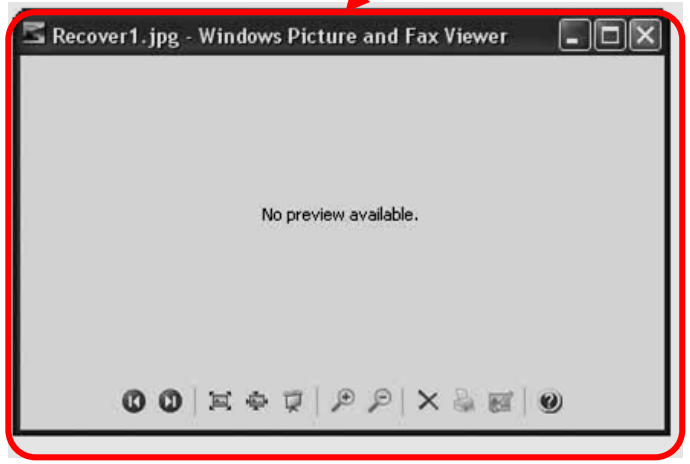
- Files labelled with *exe* extension
- Extract potential evidence

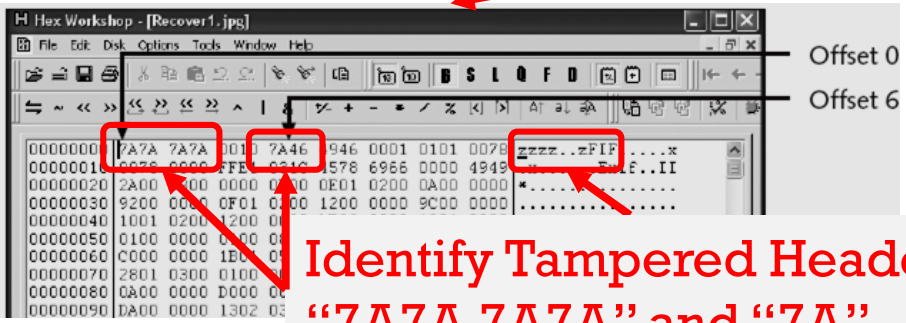
Adapted from Figure 10-11 View all clusters used by file
Adapted from Figure 10-12 Mislabelled file intentionally altered
Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

EXAMPLE: EMTS IP THEFT (REPAIR POTENTIAL EVIDENCE)

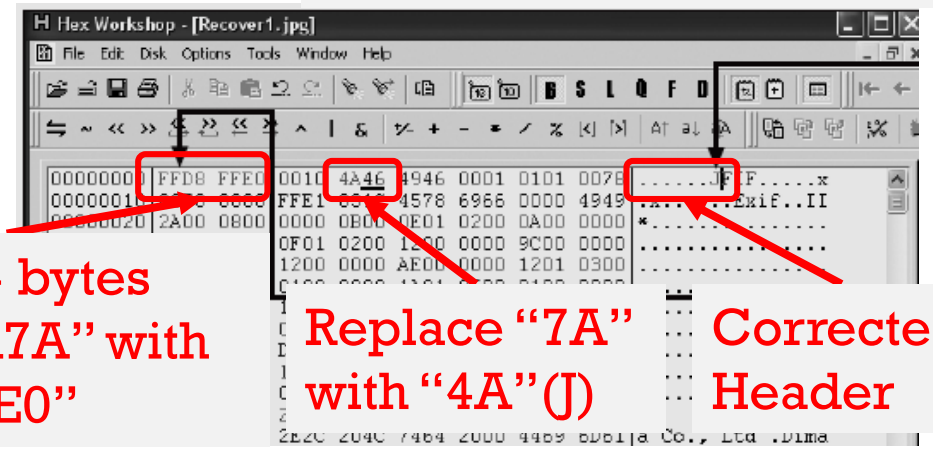
Open extracted File in Hex Editor

Try opening extracted file but unable to display content





Identify Tampered Header "7A7A 7A7A" and "7A"



Replace 4 bytes "7A7A 7A7A" with "FFD8 FFE0"

Replace "7A" with "4A"(J)

Corrected Header

Adapted from Figure 10-13 Error message of altered file
Adapted from Figure 10-14 Open altered file in Hex Editor
Adapted from Figure 10-15 Insert correct header of file
Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

EXAMPLE: EMTS IP THEFT (REPAIR POTENTIAL EVIDENCE)

ASCII hexadecimal conversion table

	0	1	2	3	4	5	6	7	8	9	A	B	C	D
0	NUL	SOH	STX	ETX	EOT	ENQ	ACK	BEL	BS	HT	LF	VT	FF	CR
1	DLE	DC1	DC2	DC3	DC4	NAK	SYN	ETB	CAN	EM	SUB	ESC	FS	GS
2	SP	!	"	#	$	%	&	'	(	)	*	+	,	.
3	0	1	2	3	4	5	6	7	8	9	:	;	<	=
4	@	A	B	C	D	E	F	G	H	I	J	K	L	M
5	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]
6	`	a	b	c	d	e	f	g	h	i	j	k	l	m
7	p	q	r	s	t	u	v	w	x	y	z	{	|	}

Second hexadecimal number

First hexadecimal number

Uppercase "A" = 41
Lowercase "a" = 61

Figure 10-16 ASCII equivalents of hexadecimal values

Adapted from Figure 10-16 ASCII equivalents of hexdecimal values
Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

EXAMPLE: EMTS IP THEFT (REPAIR POTENTIAL EVIDENCE)

Adapted from Figure 10-17 Fixed Image
Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

EXAMPLE: EMTS IP THEFT (RECONSTRUCT FROM FILE FRAGMENTS)

1. Locate the starting and ending clusters
2. For each fragmented group of clusters in the file

Locate and export all clusters of fragmented file

Determine the starting and ending cluster numbers for each fragmented group of clusters

Copy each fragmented group of clusters in their proper sequence to a recovery file

Rebuild the corrupted file's header to make it readable in a graphics viewer

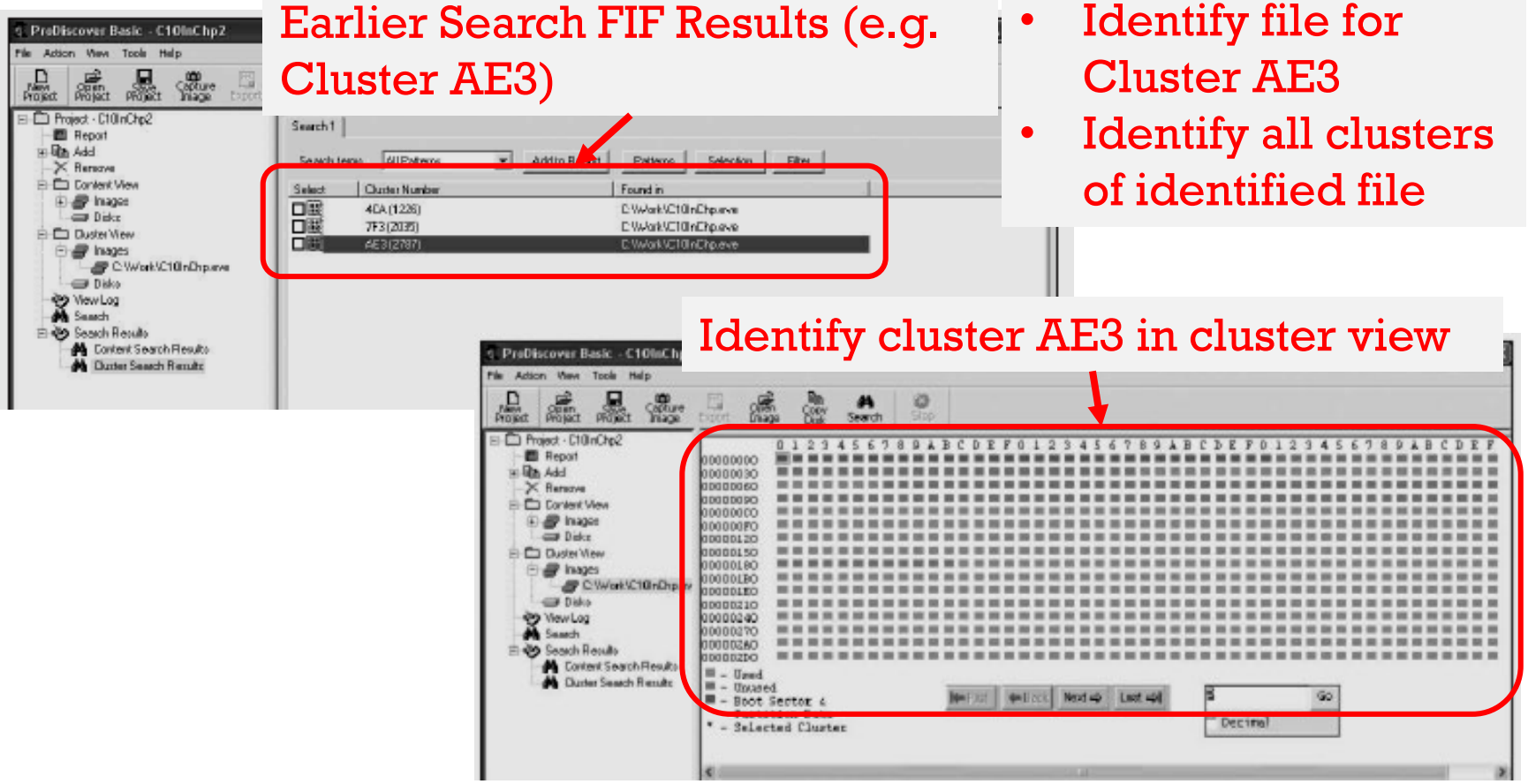
Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

EXAMPLE: EMTS IP THEFT (RECONSTRUCT FROM FILE FRAGMENTS)

Earlier Search FIF Results (e.g. Cluster AE3)

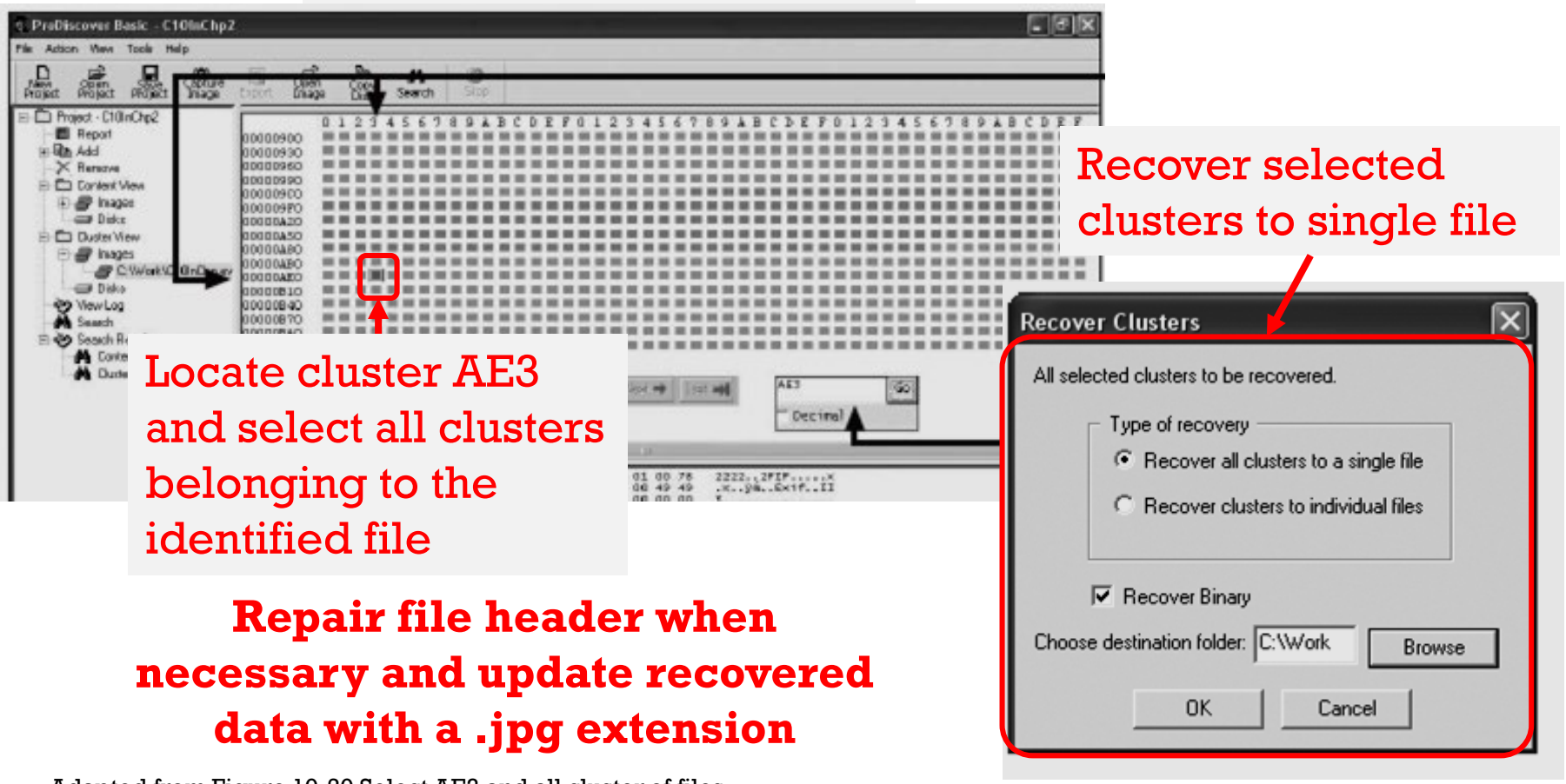
- Identify file for Cluster AE3
- Identify all clusters of identified file

Identify cluster AE3 in cluster view



Adapted from Figure 10-18 Cluster search results for AE3(2787)
Adapted from Figure 10-19 Search Cluster AE3 in Cluster View
Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

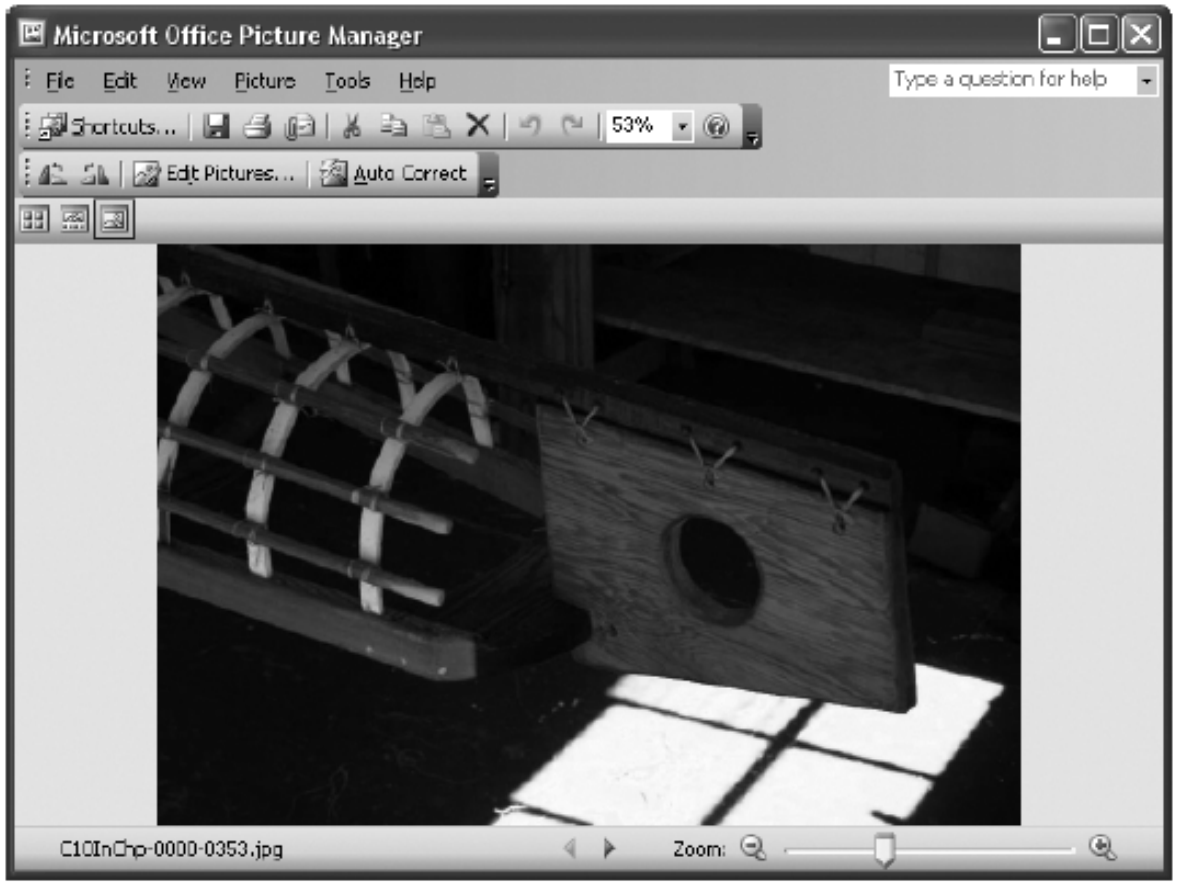
EXAMPLE: EMTS IP THEFT (RECONSTRUCT FROM FILE FRAGMENTS)



Repair file header when necessary and update recovered data with a .jpg extension

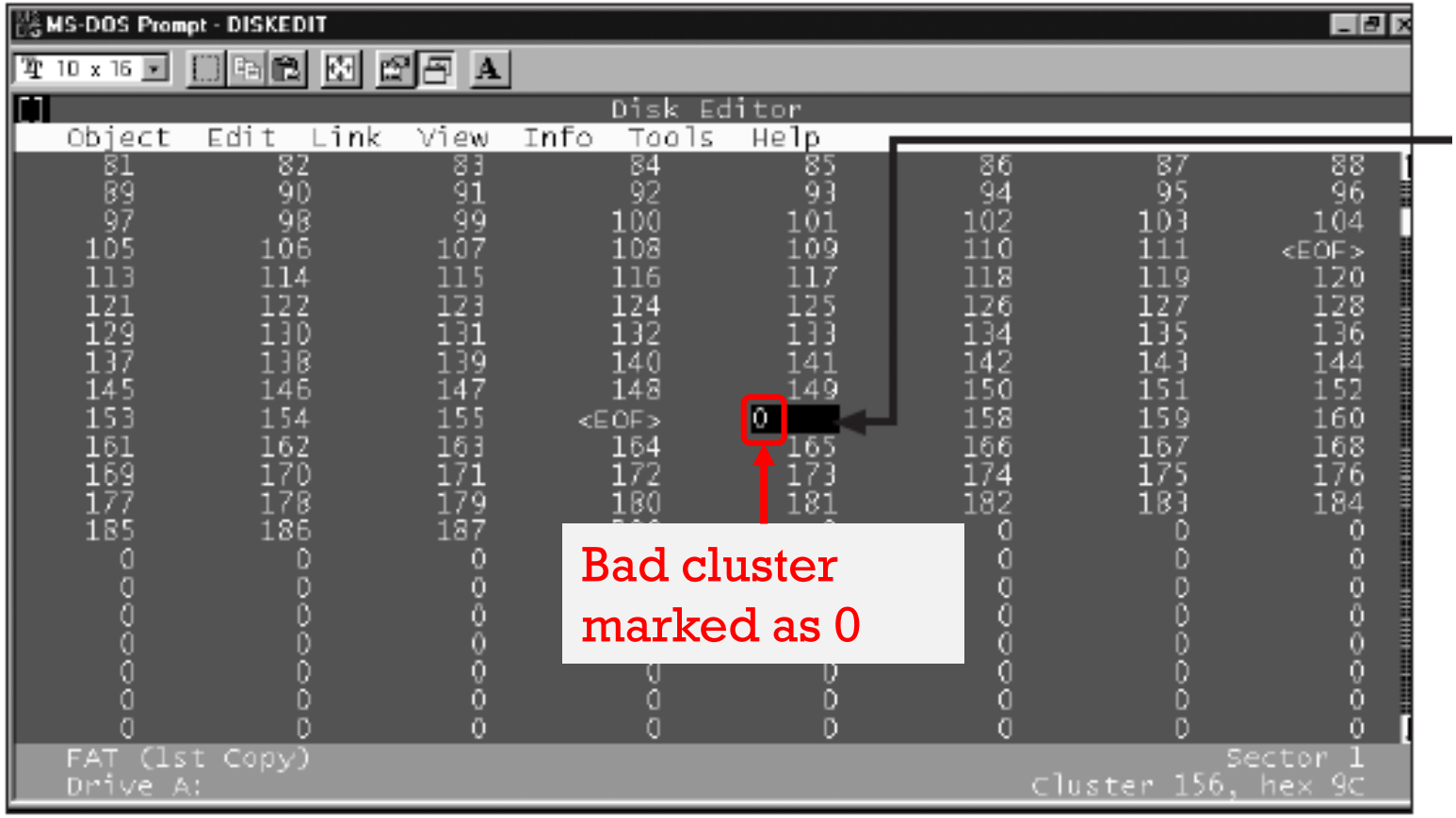
Adapted from Figure 10-20 Select AE3 and all cluster of files
Adapted from Figure 10-22 Recover all selected clusters to a single file
Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

EXAMPLE: EMTS IP THEFT (RECOVERED IMAGE FROM FILE FRAGMENTS)



Adapted from Figure 10-23 Recovered data starting from sector AE3
Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

FILE FRAGMENTS - BAD CLUSTER



Adapted from Figure 10-24 Bad Cluster marked as 0 by tools

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

IDENTIFYING UNKNOWN FILE FORMATS

What if file format is unknown?

Proprietary?

Forensic Analysis?

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- Internet is the best source
- Search engines like Google
- Find explanations and viewers
- Find files your tools do not recognize
- Use hex editor to identify hexadecimal values on header
- Use good header samples

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

ANALYZING GRAPHICS FILE HEADERS

TIFF header "4949 2A"

All TIFF files start at position zero

TIFF file headers start with hexadecimal 49, 49, 2A, which are equivalent to ASCII II

XIF header also starts with "4949 2A"

XIF header use subsequent bytes to identify XIF file

ASCII equivalent defines this file type as an extension of TIFF

Adapted from Figure 10-25 Sample header of TIFF file
Adapted from Figure 10-26 Sample Header of XIF file
Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

TOOLS FOR VIEWING IMAGES

Image Viewers	GUI forensics tools include image viewers
• ThumbsPlus • ACDSee • Quick View • IrfanView	• ProDiscover • EnCase • FTK • X-Ways Forensics • iLook • Quick View Plus

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

STEGANOGRAPHY IN FILES

Steganography → Hides information inside files

- •Ancient technique
- Hide only certain amount of information
- Usually used with image files with audio and video options
- Hard to detect

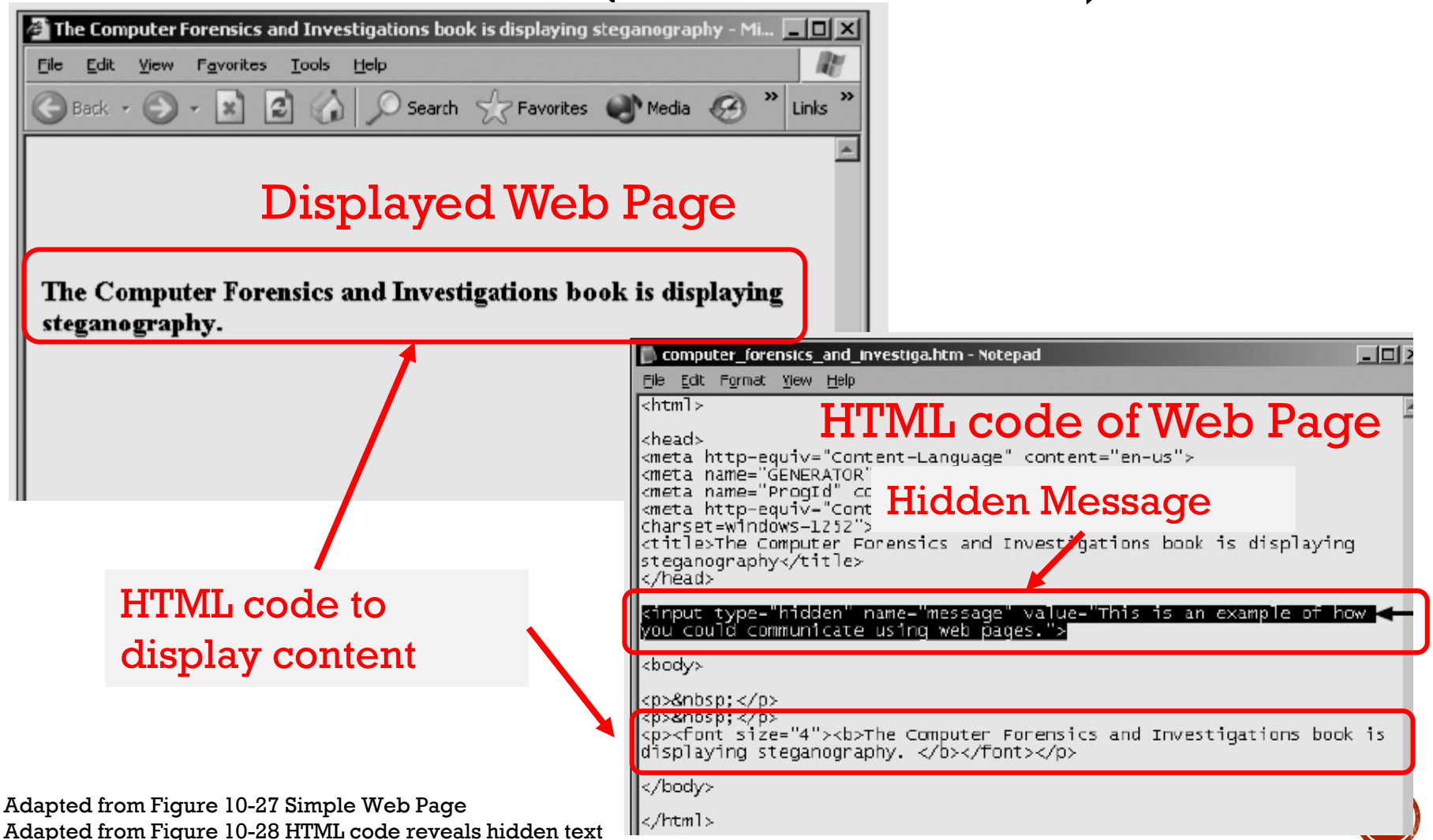
Approaches

Insertion
- Hidden data is not displayed when viewed
- Host file in its associated program to analyze the data structure

Substitution
- Replaces bits of the host file with bits of data
- Usually change the last two LSBs
- Detected with **steganalysis tools**

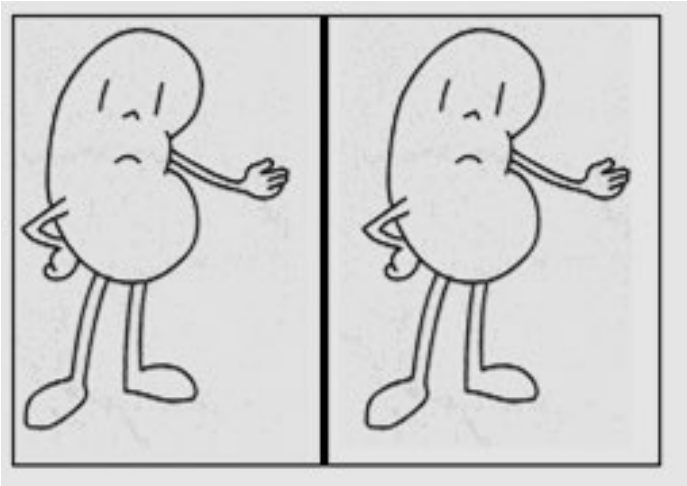
Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

STEGANOGRAPHY (HTML EXAMPLE)



Adapted from Figure 10-27 Simple Web Page
Adapted from Figure 10-28 HTML code reveals hidden text
Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

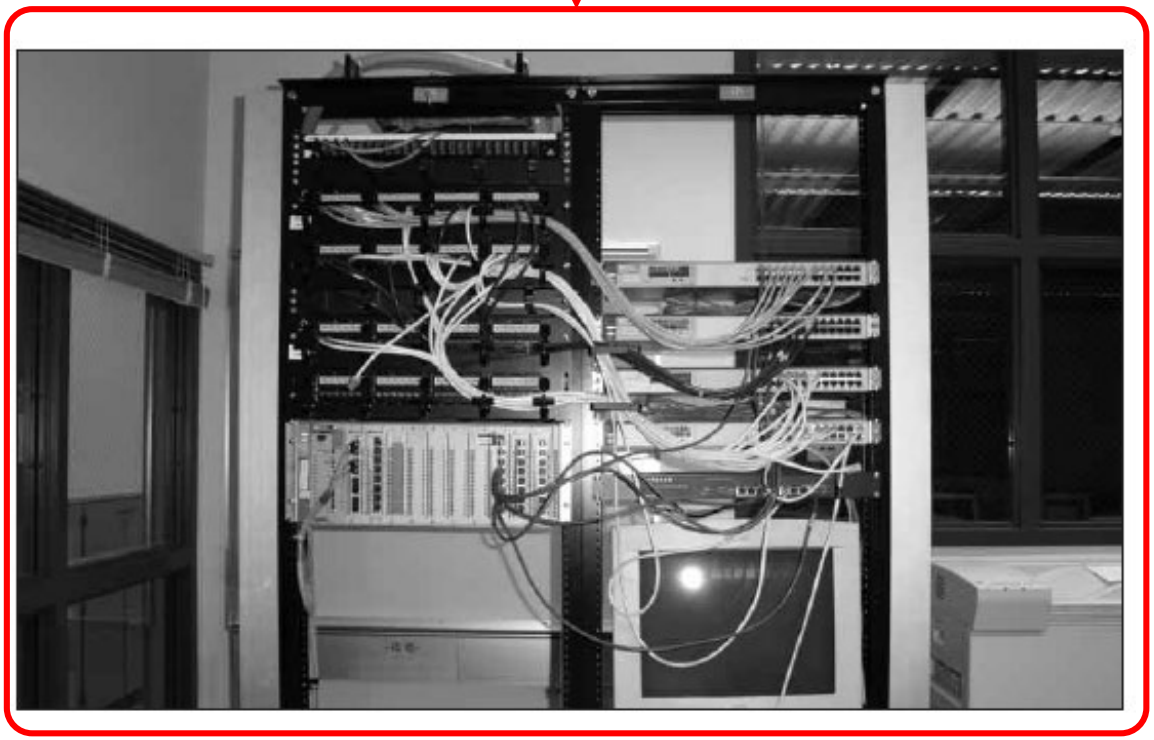
STEGANOGRAPHY (IMAGE EXAMPLE)



Original Image

Altered Image

Hidden Message in altered image



Adapted from Figure 10-29 Simple Web Page
Adapted from Figure 10-30 Hidden picture in altered image
Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

STEGANALYSIS

Steganography applied correctly → Cannot detect hidden data in most cases

Need to detect variations of the files (e.g. image/audio/video, documents, etc.)

Approaches

- Compare suspect file to good or bad image versions
- Mathematical calculations verify size and palette color
- Compare hash values

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

COPYRIGHT ISSUES

Steganography originally incorporated watermarks

Copyright laws for Internet are not clear

No International copyright law

E.g. U.S. Copyright Office - www.copyright.gov

Responsible for administering complex and dynamic set of laws

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.
https://www.copyright.gov/about/

CONCLUSION

- Image types (Bitmap/Vector/Metafile)
- Image formats (Standard/Non-Standard)
- Formats of Digital Camera images (Raw and EXIF JPEG)
- Data compression (Lossless vs Lossy)
- Recovering image files
- Image Software (Image editors/Image viewers)
- Steganography & Steganalysis

REFERENCES

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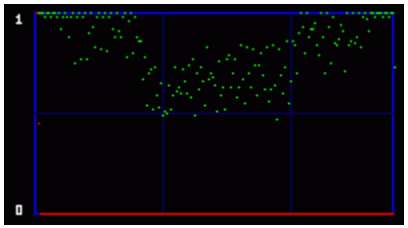
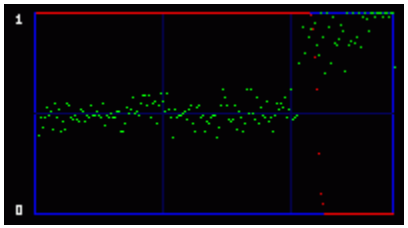
- Kleiman, D. (August 2011). The Official CHFI Study Guide (Exam 312-49). Syngress.
- Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.
- Carrier, B. (2005). *File System Forensic Analysis.* Addison-Wesley Professional.
- Messier, R. (2015). Operating System Forensics. Syngress.

EXTRA SLIDES

RESEARCH ON STEGANOGRAPHY

Hidden	Image	Enhanced LSBs	Chi-Square
Nothing	BMP	BMP	
2 kb random data	BMP	BMP